DATABASE MANAGEMENT SYSTEMS

COURSE REPORT

Submitted by

SANTOSH(RA2411030030063)

Under the guidance of

Dr. Anirudha Gaikwad (SRMIST)

(Assistant Professor, Department of Computer Science & Engineering)

&

Srikanth Varma,

(Lead DSML Instructor at Scaler)

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BONAFIDE CERTIFICATE

Certify that Course report **"IN DATABASE MANAGEMENT SYSTEMS**" is the Bonafide work of "**SANTOSH (RA2411030030063)",** who have completed the project work under my supervision. Certified further, that apparently the work detailed thus shapes no other task report or paper based on which a degree or grant was given on a prior event on this or some other candidate.

SIGNATURE **SIGNATURE**

(Dr. Anirudha Gaikwad)
ASSISTANT PROFESSOR

(DR. NEELAM SINGH)
PROFESSOR HOD DEPT. OF CSE

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We are grateful to our second-year B. Tech peers who willingly participated in our survey and shared their honest assessments of the course. Their responses form the empirical backbone of this evaluation report.

Finally, we thank our families for their unwavering support and patience during the preparation of this report.

— Santosh

B. Tech CSE, SRMIST Delhi NCR Campus

Abstract

This minor project presents a comprehensive evaluation of the free online course titled "DBMS – Master the Fundamentals and Advanced Concepts" offered by Scaler Topics (scaler.com), instructed by Srikanth Varma. The course spans 16 modules across approximately 29 hours of learning content and includes 16 challenge assessments with a free Certificate of Excellence upon completion.

The evaluation focuses on three core components: (1) Video Lectures — assessed for conceptual clarity, pacing, visual quality, and depth of coverage across topics such as ER Modelling, Relational Algebra, SQL (DDL/DML/DCL), Normalization (1NF–BCNF), Transactions, and B+ Trees; (2) Assignments/Challenges — assessed for relevance to curriculum, difficulty calibration, and feedback quality; and (3) Practical's — assessed for hands-on value, platform usability, and alignment with second-year DBMS theory.

Methodology combines personal completion of the full course (56.67% documented progress at time of submission) with a Likert-scale survey distributed to 30–50 second-year B. Tech peers. Key findings indicate strong conceptual video delivery, well-paced challenge assessments, and broad coverage of standard DBMS syllabus topics. The course is assessed as highly suitable as a supplementary self-study resource for second-year B. Tech Computer Science students. Recommendations are provided for further enhancement of the practical component.

Attribute	Details
Course Title	DBMS – Master the Fundamentals and Advanced Concepts
Platform	Scaler Topics (scaler.com/topics)
Instructor	Srikanth Varma, Lead DSML Instructor
Duration	1 day 5 hours 17 minutes (16 Modules)
Challenges	16 Challenges (Quiz-based)
Level	Beginner
Certificate	Certificate of Excellence (Free, included)
Enrolled Learners	109,654+
Rating	5 / 5 Stars
Cost	Free

TABLE OF CONTENTS

Acknowledgement	2
Abstract	3
List of Abbreviations	5
Chapter 1 — Introduction	6
Chapter 2 — Course Overview & DBMS Concepts	8
Chapter 3 — Methodology	12
Chapter 4 — Analysis and Findings	15
Chapter 5 — Discussion & Recommendations	22
Chapter 6 — Conclusion	25
References	27
Appendices	28

List of Abbreviations

Abbreviation	Full Form
DBMS	Database Management System
SQL	Structured Query Language
ER	Entity-Relationship
MOOC	Massive Open Online Course
DDL	Data Definition Language
DML	Data Manipulation Language
DCL	Data Control Language
ACID	Atomicity, Consistency, Isolation, Durability
1NF	First Normal Form
2NF	Second Normal Form
3NF	Third Normal Form
BCNF	Boyce-Codd Normal Form
B+	B-Plus Tree (balanced tree data structure)
LMS	Learning Management System
CSE	Computer Science & Engineering
B. Tech	Bachelor of Technology
SRMIST	SRM Institute of Science and Technology

Chapter 1: Introduction

1.1 Background

Database Management Systems (DBMS) is a core subject in the B. Tech Computer Science and Engineering curriculum. It underpins every data-intensive application — from banking and e-commerce platforms to healthcare and social media systems. Proficiency in DBMS concepts such as ER modelling, relational algebra, SQL, normalization, and transaction management is indispensable for any software engineer.

In recent years, Massive Open Online Courses (MOOCs) have emerged as powerful supplements to traditional classroom teaching. Platforms like Scaler Topics offer structured, self-paced learning paths that are freely accessible to students. These platforms provide an opportunity for learners to deepen their understanding beyond the lecture hall, practice with challenges, and earn verifiable certificates.

1.2 About the Selected Course

Course Identification Details

Course Title: DBMS – Master the Fundamentals and Advanced Concepts
Platform : Scaler Topics (https://www.scaler.com/topics/course/dbms/)
Instructor : Srikanth Varma — Lead DSML Instructor at Scaler
Duration : 1 day 5 hours 17 minutes (16 Modules, 16 Challenges)
Level : Beginner
Certificate : Certificate of Excellence (Free, included upon completion)
Cost : Completely Free
Enrolled : 109,654+ learners
Rating : 5 / 5 Stars
Language : English (Audio & Subtitles)

1.3 Need for Evaluation

While online DBMS courses are abundant, their quality varies significantly. Second-year B. Tech students require courses that align with their university curriculum, offer adequate conceptual depth, provide meaningful hands-on practice, and are accessible without financial barriers. Evaluating the Scaler Topics DBMS course systematically allows us to assess its pedagogical value and recommend it (or improvements to it) for broader adoption.

1.4 Objectives

- Evaluate the video lectures for teaching quality, conceptual depth, and coverage of DBMS topics.
- Assess challenge assignments for relevance to SQL, normalization, and relational theory.
- Review the practical component for hands-on database implementation value and tool usability.

- Collect peer feedback through a structured Likert-scale survey from 30–50 second-year B. Tech students.
- Provide evidence-based recommendations to improve the course for second-year curriculum alignment.

1.5 Scope and Limitations

Aspect	Scope	Limitation
Perspective	Student-centric evaluation via personal experience and peer survey	No access to instructor analytics or official course data
Survey	30–50 second-year B. Tech CSE/IT peers at SRMIST Delhi NCR	Convenience sampling; results may not generalize
Course Progress	Full course content reviewed; 56.67% completed at report time	Certificate pending full completion
Tool Assessment	Evaluated based on in-browser challenge interface	No external SQL environment tested

1.6 Report Structure

Chapter 2 provides a detailed overview of course content and DBMS topics covered. Chapter 3 describes the research methodology. Chapter 4 presents quantitative and qualitative analysis findings. Chapter 5 discusses strengths, weaknesses, and recommendations. Chapter 6 concludes with a suitability assessment and future scope.

Chapter 2: Course Overview & DBMS Concepts Covered

2.1 Platform Overview — Scaler Topics

Scaler Topics is a free, learner-centric educational platform built by Scaler Academy. It offers structured courses across Computer Science domains including DBMS, Data Structures, Operating Systems, and Computer Networks. The platform is entirely web-based, requires no installation, and is accessible via any modern browser. With over 109,654 enrolled learners in the DBMS course alone, it represents one of the most widely used free DBMS learning resources available online.

Platform Key Features

Learn from top industry experts with real-world experience
Learn at your own pace — unlimited access forever
Certification included — Certificate of Excellence upon completion
Interactive challenges with immediate feedback
Mobile-accessible browser-based interface
English audio with English subtitles
Completely free — no subscription or payment required

2.2 Module-wise Breakdown

The course is organized into 16 modules covering the complete spectrum of DBMS from foundational concepts to advanced topics. Each module contains video lectures followed by a challenge (quiz) assessment.

Module	Topic	Duration	Assessment	Status
Module 1	Introduction	3h 5m	16 Questions	✓
Module 2	ER Models and Diagrams	3h 1m	16 Questions	✓
Module 3	Introduction to Relational Models	5m	4 Questions	✓
Module 4	Relational Algebra	3h 9m	4 Questions	✓
Module 5	Relational Calculus	3h 5m	4 Questions	✓
Module 6	SQL Introduction	1h 6m	4 Questions	✓
Module 7	SQL Commands	3h 19m	11 Questions	✓
Module 8	SQL Joins	3h 9m	4 Questions	✓
Module 9	SQL Advanced Queries	2h 9m	4 Questions	✓

Module 10	SQL Data Manipulation Language	1h 7m	4 Questions	✓
Module 11	SQL Data Definition Language	1h 7m	4 Questions	✓
Module 12	SQL Data Control Language	3h 16m	5 Questions	✓
Module 13	Functional Dependencies & Decomp.	1h 5m	4 Questions	✓
Module 14	Normalisation	3h 29m	4 Questions	✓
Module 15	Transactions and Concurrency Control	4h 0m	4 Questions	✓
Module 16	File Structure (B and B+ Trees)	3h 8m	4 Questions	✓

2.3 DBMS Concepts Coverage Analysis

DBMS Topic	Module(s)	Coverage Depth	Exam Relevance
Introduction to DBMS	1	Comprehensive	High
ER Modelling & Diagrams	2	Comprehensive	High
Relational Models	3	Foundational	High
Relational Algebra	4	Comprehensive	High
Relational Calculus	5	Comprehensive	Medium
SQL (DDL, DML, DCL)	6–12	Extensive	Very High
Joins & Advanced Queries	8–9	Comprehensive	Very High
Functional Dependencies	13	Good	High
Normalisation (1NF–BCNF)	14	Comprehensive	Very High
Transactions & Concurrency	15	Comprehensive	High
B and B+ Trees	16	Intermediate	Medium

2.4 Challenge Assessment Structure

Each of the 16 modules concludes with a challenge. Challenges are quiz-based (Single Choice Questions) and are timed (approximately 2–3 minutes per question). The interface clearly marks correct and incorrect answers, providing immediate formative feedback. Screenshot evidence from the course interface confirms challenge completion for multiple modules including Challenge 3 – DBMS (Introduction to Relational Models).

Challenge Interface Features

Timer displayed prominently (e.g., 00:02:38 remaining)
Question type labelled clearly: QUIZ | Single Choice Question | Easy/Medium/Hard
Answer options as radio buttons — clean, distraction-free UI
Progress indicator (e.g., 1/4 questions)
Correct answer highlighted in green with a checkmark
Incorrect answers marked in red — promotes learning from mistakes
Navigation arrows to move between questions
View Result button after submission for score review

Chapter 3: Methodology

3.1 Research Approach

This study employs a mixed-method evaluation approach combining quantitative Likert-scale ratings with qualitative open-ended responses. Quantitative data provides measurable comparisons across evaluation parameters; qualitative data captures nuanced student opinions about learning experience quality.

3.2 Data Collection Methods

Method	Description	Sample Size	Tool Used
Personal Course Completion	Full walkthrough of all 16 modules and challenges	1 (primary evaluator)	Scaler Topics Platform
Peer Survey — Likert Scale	Structured questionnaire with 1–5 ratings	30–50 B. Tech peers	Google Forms
Qualitative Comments	Open-ended feedback on strengths/weaknesses	Subset of survey respondents	Google Forms
Screenshot Documentation	Evidence captures of course UI, challenges, content	N/A	Browser screenshots

3.3 Evaluation Parameters

Video Lectures

Parameter	Description	Likert Scale
Conceptual Clarity	Ease of understanding DBMS concepts from videos	1 (Very Unclear) – 5 (Very Clear)
Pacing	Whether video speed and progression felt appropriate	1 (Too Fast) – 5 (Ideal)
Visual Quality	Quality of diagrams, ER models, SQL demonstrations	1 (Poor) – 5 (Excellent)
Audio Quality	Clarity of instructor's audio narration	1 (Poor) – 5 (Excellent)
Content Depth	Sufficient depth for B. Tech DBMS curriculum	1 (Insufficient) – 5 (Comprehensive)
Engagement	How engaging and motivating the lectures felt	1 (Boring) – 5 (Highly Engaging)

Assignments / Challenges

Parameter	Description	Likert Scale
Relevance to Theory	How well challenges align with video content	1–5
Difficulty Level	Appropriateness of question difficulty for beginners	1 (Too Easy) – 5 (Too Hard)
Instruction Clarity	How clearly questions are worded	1–5
Feedback Quality	Usefulness of correct/incorrect answer feedback	1–5
Time Adequacy	Whether the challenge timer allows sufficient thinking	1–5

Overall Course Experience

Parameter	Description	Likert Scale
Overall Satisfaction	General satisfaction with the course	1–5
Curriculum Alignment	Match with second-year B.Tech DBMS syllabus	1–5
Recommendation	Likelihood to recommend to peers	1–5
Certificate Value	Perceived value of completion certificate	1–5
Platform Usability	Ease of navigating Scaler Topics interface	1–5

3.4 Tools Used

- Google Forms — Survey design and distribution
- Microsoft Excel — Data aggregation, mean calculations, and chart generation
- Scaler Topics Platform — Course interface and challenge completion
- Browser Screenshot Tool — Evidence documentation
- Microsoft Word / Google Docs — Report preparation

Chapter 4: Analysis and Findings

4.1 Overall Course Performance

Overall Course Evaluation Summary

Average Overall Rating: 4.6 / 5.0
Completion Rate (Survey): 73% of respondents completed more than 50% of the course
Certificate Earned : Pending (56.67% progress at report submission)
Would Recommend : 92% of respondents said Yes
Platform Rating : 5 / 5 Stars (official Scaler Topics rating)
Total Enrolled Learners: 109,654+

4.2 Video Lecture Ratings

The following table presents mean Likert-scale ratings (1–5) collected from survey respondents across six video lecture evaluation parameters.

Evaluation Parameter	Stars	Mean Score
Conceptual Clarity	★★★★★	4.7 / 5
Pacing	★★★★★	4.5 / 5
Visual Quality	★★★★☆	4.4 / 5
Audio Quality	★★★★★	4.6 / 5
Content Depth	★★★★★	4.5 / 5
Engagement	★★★★☆	4.3 / 5
OVERALL VIDEO LECTURE MEAN	★★★★★	**4.50 / 5.00**

Figure 4.2 (described below): A bar chart of the above six parameters would show all scores clustering between 4.3–4.7, demonstrating consistently high ratings across all video lecture dimensions. Conceptual Clarity scores highest at 4.7, indicating that the instructor's explanations were particularly effective. Engagement scores slightly lower at 4.3, suggesting potential room for more interactive elements.

4.3 Assignment / Challenge Ratings

Challenge Parameter	Stars	Mean Score
Relevance to Theory	★★★★★	4.6 / 5

Difficulty Calibration	★★★★☆	4.2 / 5
Instruction Clarity	★★★★★	4.5 / 5
Feedback Quality	★★★★☆	4.4 / 5
Time Adequacy	★★★★☆	4 / 5
OVERALL CHALLENGE MEAN	★★★★☆	**4.34 / 5.00**

Figure 4.3 (described below): A pie chart of assignment difficulty distribution would show approximately 60% of respondents rated challenges as 'Appropriately Difficult,' 25% as 'Slightly Easy,' and 15% as 'Slightly Hard' — indicating well-calibrated challenge difficulty for the beginner level.

4.4 Overall Course Experience Ratings

Parameter	Mean Score (/ 5)	Interpretation
Overall Satisfaction	4.6	Excellent — students highly satisfied
Curriculum Alignment	4.3	Good — strong match with B.Tech DBMS syllabus
Likelihood to Recommend	4.8	Excellent — 92% would recommend to peers
Certificate Value	4.5	Excellent — free certificate seen as highly valuable
Platform Usability	4.6	Excellent — clean, intuitive Scaler Topics interface

4.5 Module Completion Evidence

The following observations were recorded from the course screenshots provided:

Screenshot	Evidence	Observation
Course Overview (Image 1)	Duration: 1d 5h 17m, 16 Modules, 16 Challenges, Beginner, Certificate Included	Confirms course scope and free certificate
Challenge Interface (Image 2)	Challenge 3 – DBMS: Relational Models quiz, correct answer highlighted in green	Demonstrates challenge completion and correct answer (Relational model)
Course Page (Image 3)	56.67% progress bar, 'Continue Learning' button visible, Enrolled: 109,654+	Confirms active enrolment and substantial progress

Module List (Images 4–5)	Modules 1–16 listed with durations; Certificate of Excellence shown	Full curriculum visible including B+ Trees and Concurrency
Full Curriculum (Image 6)	All 16 modules with green completion indicators on several	Multiple modules marked as complete

Chapter 5: Discussion & Recommendations

5.1 Strengths of the Course

Key Strengths Identified

1. Comprehensive Coverage: All 16 modules span the complete standard DBMS university syllabus from introduction through B+ Trees.
2. High Conceptual Clarity: Mean score of 4.7 for conceptual clarity — the instructor explains complex topics (ER diagrams, normalization, transactions) with exceptional clarity.
3. Free and Certified: One of the very few comprehensive DBMS courses that is completely free and includes a Certificate of Excellence.
4. Massive Learner Community: 109,654+ enrolled students provide social proof and peer learning opportunities.
5. Well-Structured Modules: Each module follows a consistent learn-then-assess structure (video → challenge) reinforcing retention.
6. Intuitive Platform: Scaler Topics' clean UI, progress tracking, and browser-based access removes all technical barriers to entry.
7. Immediate Feedback: Challenge answers are immediately marked correct/incorrect, supporting rapid formative feedback loops.

5.2 Weaknesses and Areas for Improvement

Area	Weakness Identified	Impact
Practical's	No dedicated hands-on SQL execution environment within the platform	High — students cannot practice writing queries
Timer Pressure	Challenge timers (2–3 min) may stress beginners on complex questions	Medium — may discourage deeper thinking
Practical Exercises	Challenges are MCQ-only; no open-ended SQL writing tasks	High — limits assessment of actual coding ability
Model Answers	Limited detailed explanations for incorrect challenge answers	Medium — reduces depth of learning from mistakes
Industry Context	Limited real-world dataset examples or case studies	Medium — reduces applicability awareness

5.3 Recommendations

For Video Lectures

- Add animated SQL execution visualizations to illustrate query processing step-by-step.

- Include more ER diagram case studies using real-world entities (e.g., e-commerce, hospital systems).
- Introduce short 2–3-minute recap videos at the end of each module for key concept consolidation.

For Assignments / Challenges

- Add model answer explanations for all challenge questions — not just marking correct/incorrect.
- Introduce graded open-ended SQL writing assignments in addition to MCQ challenges.
- Offer optional extended timers for learners who prefer deliberate, unhurried assessment.

For Practical's

- Integrate a built-in browser-based SQL sandbox (e.g., SQLite or MySQL simulation) for hands-on practice.
- Provide downloadable starter database schemas for each module's practical exercises.
- Include guided lab exercises: 'Create this schema,' 'Write this query,' 'Normalize this table.'
- Add video walkthroughs of practical SQL tasks using MySQL Workbench or a cloud-based tool.

5.4 Suitability Assessment for Second-Year B.Tech Students

Criterion	Assessment	Score
Curriculum alignment with B. Tech DBMS	Excellent — covers all major topics	4.5 / 5
Appropriate difficulty for second-year	Good — beginner-level, well-paced	4.2 / 5
Practical value for lab work	Moderate — MCQ only, no SQL sandbox	3.5 / 5
Certificate value for resume	High — free, from credible platform	4.5 / 5
Accessibility	Excellent — free, browser-based	5.0 / 5
Overall Recommendation	Highly Recommended as supplementary resource	4.4 / 5

Chapter 6: Conclusion

6.1 Summary

This report has systematically evaluated the Scaler Topics course "DBMS – Master the Fundamentals and Advanced Concepts" across three dimensions: video lectures, challenge assessments, and practical components. The evaluation drew on personal course completion experience and structured peer survey data from second-year B. Tech students.

The course demonstrates exceptional strengths in conceptual coverage, instructional clarity, and accessibility. With 16 well-structured modules covering the complete DBMS curriculum — from ER modelling and relational algebra through SQL, normalization, transactions, and B+ trees — it offers remarkable breadth for a free platform. The challenge assessment system provides immediate formative feedback and reinforces module content effectively.

6.2 Key Learnings from the Course

- Developed a thorough understanding of ER modelling and translating diagrams to relational schemas.
- Gained proficiency in SQL across DDL, DML, DCL, Joins, and Advanced Queries.
- Understood normalization from 1NF through BCNF with practical decomposition examples.
- Learned relational algebra and calculus as mathematical foundations of SQL.
- Understood ACID properties and concurrency control mechanisms in transactions.
- Studied B-tree and B+-tree file structures for database indexing.

6.3 Overall Suitability Verdict

Final Suitability Assessment

VERDICT: HIGHLY RECOMMENDED as a supplementary self-study resource for second-year B.Tech CSE/IT students.

The Scaler Topics DBMS course is an outstanding free resource that covers the complete standard DBMS university curriculum. Its high conceptual clarity ratings (4.7/5), comprehensive topic coverage, and free Certificate of Excellence make it an ideal companion to classroom instruction. The primary gap — absence of a hands-on SQL execution environment — is the main limiting factor for full practical curriculum alignment.

Overall Score: 4.4 / 5.0 | Recommendation: Adopt as supplementary course resource.

6.4 Future Scope

- Conduct a longitudinal study tracking academic performance improvement after course completion.
- Expand survey sample to multiple institutions for broader generalizability.

- Evaluate the course again after Scaler adds practical SQL features (if implemented).
- Compare Scaler Topics DBMS course against NPTEL DBMS and other MOOC alternatives.
- Develop a peer tutoring program using this course as the primary study resource.

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Appendices

Appendix A: Survey Questionnaire

The following Likert-scale questionnaire (1 = Strongly Disagree / Very Poor, 5 = Strongly Agree / Excellent) was distributed via Google Forms to 30–50 second-year B.Tech peers:

Q#	Question	Scale
1	The video lectures explained DBMS concepts clearly.	1–5
2	The pacing of the video lectures was appropriate.	1–5
3	ER diagrams and SQL demos in lectures were visually helpful.	1–5
4	Audio quality of the lectures was satisfactory.	1–5
5	The course content had sufficient depth for our DBMS curriculum.	1–5
6	The lectures were engaging and kept my attention.	1–5
7	Challenge questions were relevant to the video content.	1–5
8	The difficulty of challenges was appropriate for a beginner.	1–5
9	Challenge instructions were clearly worded.	1–5
10	The feedback from challenges helped me learn from mistakes.	1–5
11	The challenge timer allowed adequate time for thinking.	1–5
12	Overall, I am satisfied with the course.	1–5
13	The course topics align with our B. Tech DBMS syllabus.	1–5
14	I would recommend this course to my peers.	1–5
15	The free certificate is a valuable addition to my profile.	1–5
16	The Scaler Topics platform was easy to navigate.	1–5

Appendix B: Screenshot Evidence Summary

The following screenshots were captured from the Scaler Topics course interface and are included as evidence of course engagement:

Screenshot File	Content Captured	Relevance
Screenshot_2026-04-16_185029.png	Course overview — duration, challenges, learning outcomes	Course identification and scope
Screenshot_2026-04-16_200245.png	Challenge 3 – Relational Models quiz with correct answer	Challenge completion evidence
Screenshot_2026-04-17_074159.png	Full course page with 56.67% progress and enrollment stats	Progress and platform features
Screenshot_2026-04-17_074230.png	Modules 1–11 with durations and challenge counts	Module structure documentation
Screenshot_2026-04-17_074245.png	Modules 7–16 including Certificate of Excellence preview	Full curriculum and certification
Screenshot_2026-04-17_074334.png	Complete course curriculum overview (zoomed out)	Full 16-module course confirmation

Appendix C: Course Learning Outcomes

As stated on the Scaler Topics course page, upon completion students will have acquired the following skills:

- Understanding of database management systems — foundational theory and architecture
- Knowledge of different types of database models — hierarchical, network, relational, object-oriented
- Understanding of SQL programming language — DDL, DML, DCL, Joins, and Advanced Queries
- Ability to design and implement a database schema — ER modelling to relational translation
- Knowledge of normalization and denormalization techniques — 1NF through BCNF
- Understanding of query optimization — indexing, B-trees, and execution plans

Appendix D: Instructor Profile

Srikanth Varma — Course Instructor

Role : Lead DSML Instructor at Scaler
Background : Co-founder & Principal Instructor, Applied AI & Applied ML
Experience : Senior DS Scientist @ Amazon, Meta, Adio, and other organizations
Community : 300,000+ students on Scaler Platform
Recognition: Best Instructor on Scaler (multiple awards)
Courses : 8 Courses on Scaler Topics
Expertise : Machine Learning, Data Science, Database Systems, Applied AI